

IMO 2026 — Problem 5

Solution by xAI Grok 4.5

2026-07-21

Source: `results/grok-4.5/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round run: earlier-round audit logs in `scratch/`.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

partial

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Approaches tried

- Specialization and classical inequalities (RMS-AM-GM): when $x = f(y)$ both outer terms equal x , forcing $(f(x) + y)/2 = x$ whenever $y \in f^{-1}(x)$. Outer comparison always true by $(x - f(y))^2 \geq 0$. When $x = y$ the ineq is automatic for any f .
- Tested $f(x) = cx$: right inequality always holds; left holds for all $x, y > 0$ iff $c = 1$. Verified algebraically (quadratic form discriminant $8(c^2 - 1)^2$) and numerically.
- Constants and c/x fail (growth mismatch).
- Next: prove surjectivity of f (or characterize image), then deduce $f(2x - f(x)) = x$ on preimages, or prove $f(x) = x$ directly by other substitutions / taking infima.

Current best

$f(x) = x$ works. It is the unique positive multiple of identity. Strong structural constraint: if $a \in \text{im}(f)$ and $f(y) = a$ then necessarily $y = 2a - f(a)$ (and this must be positive). Aim to prove $f = \text{id}$ is the only solution.

Full proof

(empty)